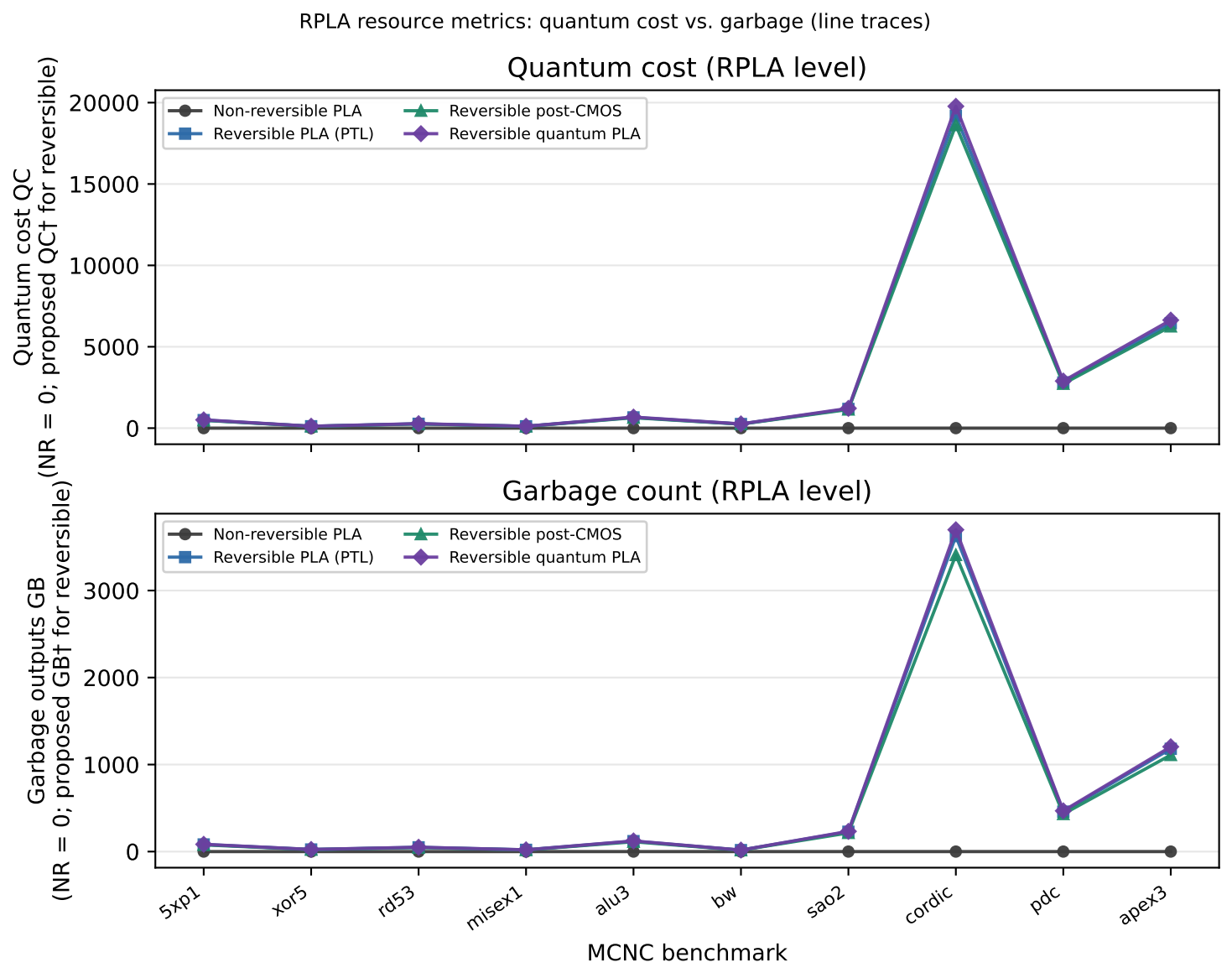
RPLA resource metrics: quantum cost vs. garbage (line traces)
Quantum cost (RPLA level)
Quantum cost QC
(NR = 0; proposed QC† for reversible)
Non-reversible PLA
Reversible PLA (PTL)
Reversible post-CMOS
Reversible quantum PLA
20000
15000
10000
5000
0
Garbage count (RPLA level)
Garbage outputs GB
(NR = 0; proposed GB† for reversible)
3000
2000
1000
0
5xp1
xor5
rd53
misex1
alu3
bw
sao2
cordic
pdc
apex3
MCNC benchmark